Mattia Fait

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Education

BSc in Electronic Engineering — *Trento, Italy*
University of Trento — Present

- Thesis: Working on integrated photonics.
- Focus: Photonics, Digital Logic Design, and Signal Processing.

Selected Public Projects

nixos-config — *Systems & Infrastructure*
NixOS workstation and local NAS configuration — Present

- Declarative system configuration for a high-efficiency workstation and local NAS, including Hyprland, networking, storage, containers, development tools, and reproducible operations workflows.
- **Stack:** NixOS, Nix flakes, Bash, Hyprland, Docker, Samba, Syncthing.

RISC-V — *Computer Architecture*
Synthesizable RV32IM/Zicsr CPU core in VHDL — Present

- Designed a modular single-cycle and five-stage pipelined RISC-V core with hazards, forwarding, interrupts, CSRs, memory-mapped peripherals, simulation, and FPGA synthesis flows.
- **Stack:** VHDL, GHDL, Vivado, RISC-V, FPGA, Make.

ppse-lab-firmware — *Embedded Systems*
RP2040 GPS and environmental telemetry logger — Present

- Implemented real-time GPS acquisition, telemetry, persistent logging, USB mass storage, OLED UI, and sensor feedback for an RP2040-based embedded system.
- **Stack:** C/C++, RP2040, FreeRTOS, TinyUSB, LittleFS, GNSS, SSD1306.

ppse-hardware — *Digital Hardware*
Embedded-system PCB and electronics design — Present

- Designed the KiCad schematics and PCB for the companion embedded hardware, integrating the microcontroller, power, user interface, and sensor subsystems.
- **Stack:** KiCad, PCB design, electronics, embedded hardware.

ald-cgra — *Computer Architecture*
Runtime-reconfigurable coarse-grained array on FPGA — Present

- Developed a 4x4 processing-element array with a UART-controlled host tool, emulator, custom configuration language, simulation, synthesis, and verification flows.
- **Stack:** VHDL, C, FPGA, UART, GHDL, Vivado, Make.

synthesis — *Formal Methods*
Lean 4 formal engineering frontend and semantic IR — Present

- Built an extensible formal framework for typed engineering models, requirements, refinement, domain theories, and machine-checkable proofs.
- **Stack:** Lean 4, Mathlib, formal verification, semantic IR, Nix.

retro-games — *Systems Programming*
Terminal-based classic games collection — Present

- Implemented a modular ncurses game suite including Snake, Tetris, Pac-Man, Pong, Breakout, Chess, Minesweeper, and Sudoku.
- **Stack:** C, ncurses, GNU Autotools, Unix, modular systems programming.

Technical Skills

Formal Verification	Lean 4, Formal Logic, Metaprogramming
Languages	C, C++, Rust, Python, Bash, LaTeX
Systems	Arch Linux (custom ISO), NixOS, Docker
DevOps/Tools	Git, CI/CD, Neovim (custom config), Makefile
Electronics	Embedded Systems, PCB Design (KiCad), Photonics

LANGUAGES

Italian	Native
English	Upper-Intermediate (B2)
	Intensive B2 Immersion Program – Oxford, UK (Summer 2018)
German	Elementary (A2); 8 years of school study